SOP Number and version: SOP-NAT-AST.v01

SOP Title: **Antimicrobial Susceptibility Testing**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
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Approval	Frank J. Hernandez	Principal investigator		09/01/2025

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1. Overview and background

Antimicrobial susceptibility testing (AST) is a crucial step of antibiotic development. In Nucleic Acids Technology Laboratories, gold standard method in AST, broth microdilution method (BMD) is used. The main objective of the work is to investigate the minimum drug concentration that inhibits the growth of tested bacteria (minimum inhibitory concentration; MIC).

2. Scope

In this SOP, antimicrobial susceptibility testing methods used in Nucleic Acids Technology Laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686) will be explained.

This SOP covers the sections below.

- Drug preparation
- BMD plate preparation and incubation
- Evaluations of the results

3. Procedure

3.1. Drug preparation

- Prepare the solvent to be used to dissolve the drug according to the manufacturer`s recommendations. Autoclave, if needed.

- Stock concentration of the drug must be at least two times higher than the highest drug concentration to be tested. Weight the drug accordingly using a high precision analytical lab scale **(Figure 1)** and plastic weighing boats **(Figure 2).**
- Mix the drug with solvent and vortex until it dissolves.
- Store the stock solution at -20°C.

Figure 1. Lab scale

Figure 2. Weighing boats

3.2. BMD plate preparation

- Use a 96-well U-bottom polypropylene plate with lid **(Figure 3**).
- Always have at least three media control (no bacteria) and three growth control (no drug) wells in the plate design (e.g., A1-B1-C1 for media control and A12-B12-C12 for growth control).
- Add 50 µL PBS (buffer can be changed based on the requirements of the experiment) in each well except the wells allocated for highest drug concentration (e.g., A2-B2-C2).
- Add 100 µL drug (x2 starting concentration) in the wells allocated for highest drug concentration (e.g., A2-B2-C2).
- Perform the serial dilutions by transferring 50 µL drug to the following wells (e.g., A3-B3-C3 → A4-B4-C4 → A5-B5-C5…) until the last wells before the growth control wells (e.g., A11-B11-C11).
- After the last transfer (e.g., to A11-B11-C11), discard the remaining 50 µL drug.
- **Note!:** While performing serial dilutions, cover the growth control wells (e.g., A12-B12-C12) using the lid to reduce the possibility of making mistakes by adding drug to those wells!
- Prepare 10^6 CFU/mL bacteria solution in cation adjusted Mueller Hinton Broth.
- For this purpose, transfer 1-2 colonies into a sterile 16 mm glass tube with 2 mL PBS using a sterile inoculation loop and vortex.
- Use McFarland Densitometer **(Figure 4)** to obtain a bacterial suspension of 0.5 McFarland which equals 1.5×10^8 CFU/mL.
- Dilute this suspension 1/150 using cation adjusted Mueller Hinton Broth to reach 10^6 CFU/mL.

- Distribute 50 µL bacteria solution into each well except the media control wells (e.g., A1-B1-C1). This will make the final bacteria concentration $5x10^5$ CFU/mL.
- Close the lid and incubate the plate at 36±1°C for 18 hours.
- If incubation takes place in a plate reader **(Figure 5)** according to the experimental design, use a protocol to measure OD at 600nm for every 10 minutes to obtain the growth curves at the end of the experiment.
- All bacteria/drug concentration combinations should be tested in triplicates.

Figure 3. 96-well U-bottom plate with lid

Figure 4. McFarland Densitometer

Figure 5. Plate Reader

Figure 6. Example of growth curves based on OD measurements over 18 hours.

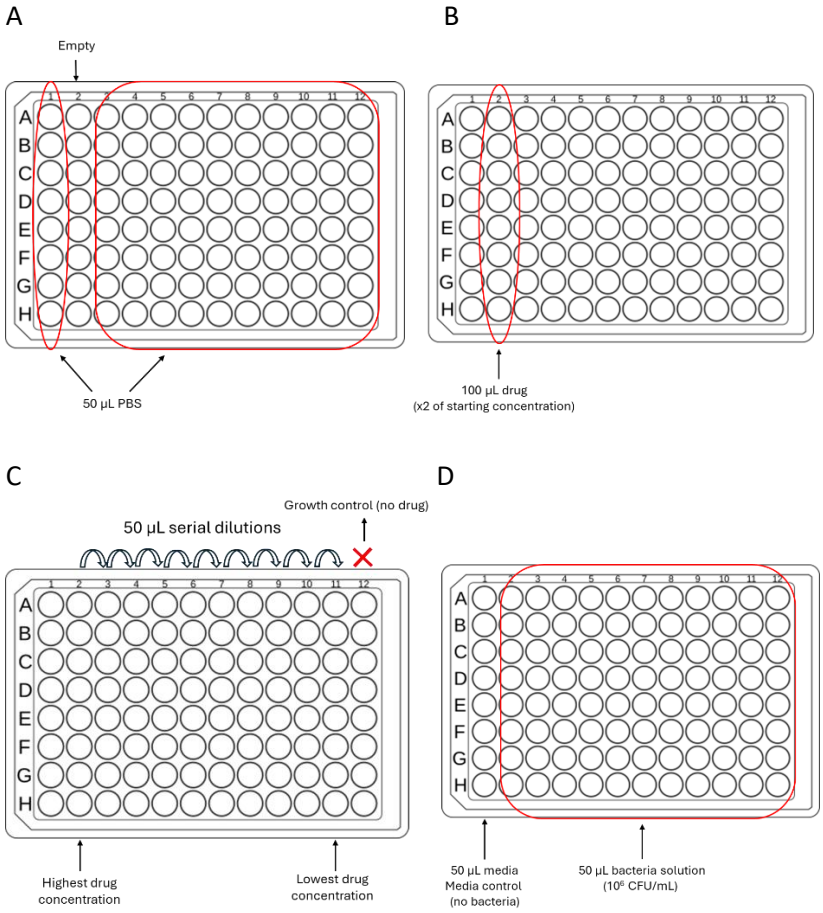

Figure 7. Steps of BMD plate preparation.

A) Adding 50 µL PBS all the wells except the wells allocated for highest drug concentration.
B) Add 100 µL drug (x2 starting concentration) in the wells allocated for highest drug concentration.
C) Perform the serial dilutions by transferring 50 µL drug to the following wells until the last wells before the growth control wells.
D) Distribute 50 µL bacteria solution into each well except the media control wells.

3.3. Evaluation of the results

- Before evaluating the results, make sure that there is no growth in media control wells and full bacterial growth in growth control wells.
- MIC value is defined as the lowest drug concentration in the well that bacterial growth cannot be observed with a naked eye.
- Export the raw data from the plate reader.
- Calculate the average and SD values of triplicated wells for each time point.

- Draw the growth curves using these averages.
- Report the results as described in the **SOP-NAT-DATA.**

4. Purpose/Objective

The objective of this SOP is to guide Nucleic Acids Technology staff on performing antimicrobial susceptibility testing in Nucleic Acids Technology Laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- R&D – Research and Development
- CTO – Chief Technology Officer
- CSO – Chief Scientific Officer
- COO – Chief Operations Officer
- PPE – Personal Protective Equipment
- BSL-2 – Biosafety Level 2
- AST – Antimicrobial Susceptibility Testing
- BMD – broth microdilution method
- MIC – Minimum Inhibitory Concentration
- CFU – Colony Forming Unit
- SD – Standard Deviation
- OD – Optical Density
- PBS – Phosphate Buffered Saline
- µL – Microliter
- mL – Milliliter

6. Responsibilities

This SOP was prepared by Student name, revised by Baris Ata Borsa and approved by Frank J. Hernandez, principal investigator.

7. Equipment/Material

- High precision analytical balance
- Metal measuring spoons
- Weighing boats
- BSL-2 biosafety cabinet
- Inoculation loops
- Bunsen burner
- McFarland densitometer
- 96-well U-bottom plates with lid
- 16 mm glass tubes

- Gloves
- Pipettes
- Pipette tips
- Incubator
- Plate reader

8. Caution/Warning

- Use BSL-2 biosafety cabinet for all the work that involves bacteria.
- Use PPE according to **SOP-NAT-BSL2**.
- Clean the working area before and after the work.

9. References

European Committee on Antimicrobial Susceptibility Testing (EUCAST). MIC determination of non-fastidious and fastidious organisms. (https://www.eucast.org/ast_of_bacteria/mic_determination)

10. Version Control

Original version of the **SOP-NAT-AST.v01**Check **Table 1**.

11.Attachments/Appendices

Appendix I:

- Bunsen burner manual
- McFarland Densitometer manual
- Plate reader manual
- Lab scale manual
- BSL-2 biosafety cabinet manual
- Incubator manual